Hash Game Studio

Cube World Puzzle – Starter kit

User Guide

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Introduction

This system will allow you to create functional cube shape map easily for cube puzzle games and do it really easy. Just click over menu item named ‘Cube Worlds puzzles’ and create terrain object, create your own map that AI helps you many times, define hero or heroes and their goal homes. The many tools designed for this purpose, which will be explained below, will also be of great help to you. Create your custom map in minimum time and make your game logic. Just enjoy it.

This system is designed in such a way that the player must place cube(s) called ‘Hero’ in their original positions called ‘Goal home’. For this purpose, mechanisms have been designed to design diverse and endless stages. Among the most important strengths of this package are: AI in helping to produce endless environmental maps in the game, optimizing the game`s environmental space, and at the same time, their simplicity.

Scripts list

There you have the most important scripts to work.

Table 1-Scripts

Hero			
HeroController.cs	InputController.cs	HeroLineCensorController.cs	InventoryController.cs
Terrain			
TerrainManager.cs	Obstacle.cs	Door.cs	InformerHandler.cs
AutoWalk.cs	AutoWalkDestination.cs	CheckPoint.cs	DestructibleObstacle.cs
DoorLever.cs	GoalHome.cs	Portal.cs	PortalDestination.cs
Spawner.cs	SpawnerButton.cs	Undercover.cs	TerrainData.cs
ObstacleTypesBase.cs	CollectableObject.cs	IsometricCameraController .cs	KeyController.cs
Path algorithms and classes			
DungeonGenerator.cs	RandomWalk	TerrainLand.cs	Quadrilateral.cs
SnakePath			
Scriptable object			
InputControllerSettings.cs	HeroSettings.cs	HeroAudioSettings.cs	TerrainAudioSettings.cs
TerrainEditorSettings.cs	TerrainLayout.cs	AutoWalkSettings.cs	DestructibleSettings.cs
SpawnerSettings.cs			
Editor			
Any classes that ended with Editor.cs			
Functional and helper classes			
Any classes located in the ‘Core’ folder.			

Note:
You don`t have to know each script or programming. This list is just for awareness.

How to use menu?

Here the menu items explained quickly but in scripts title, it explained with more details.

Terrain

Terrain is the main controller in this system. All obstacles' details are storing and managing in this part. All audio clips and win event are handling from this part.

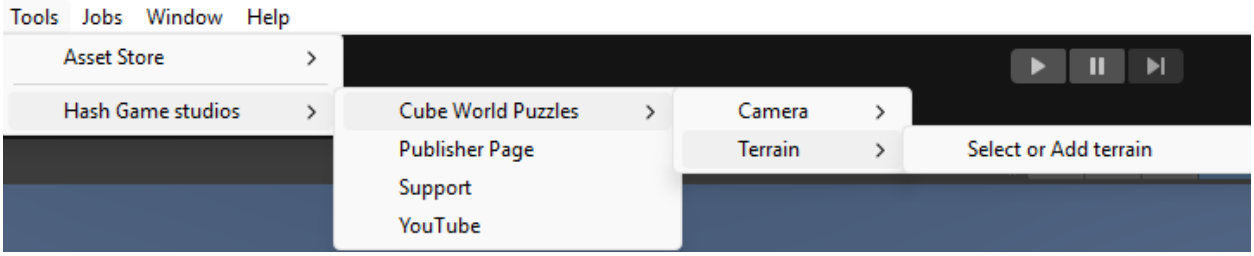

Figure 1-1Terrain menu

Camera

This part is attaching the isometric camera script to the main camera and select it after this action. Other custom scripts can also be used.

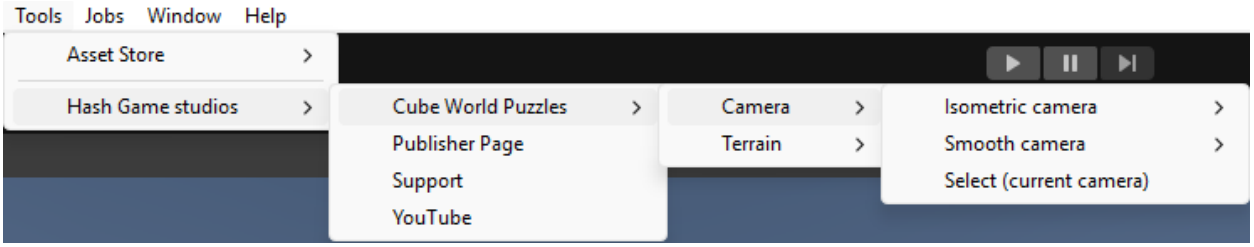

Figure 2-Camera menu

Important scripts and parameters

Here are the most important and useful scripts for game design explained.

Scriptable objects

Scriptable objects are pre-created by default at the following address. If needed, new settings can be created and used by right-clicking on the Unity Editor environment.

Path: Cube Worlds Puzzles\Recourses\Settings

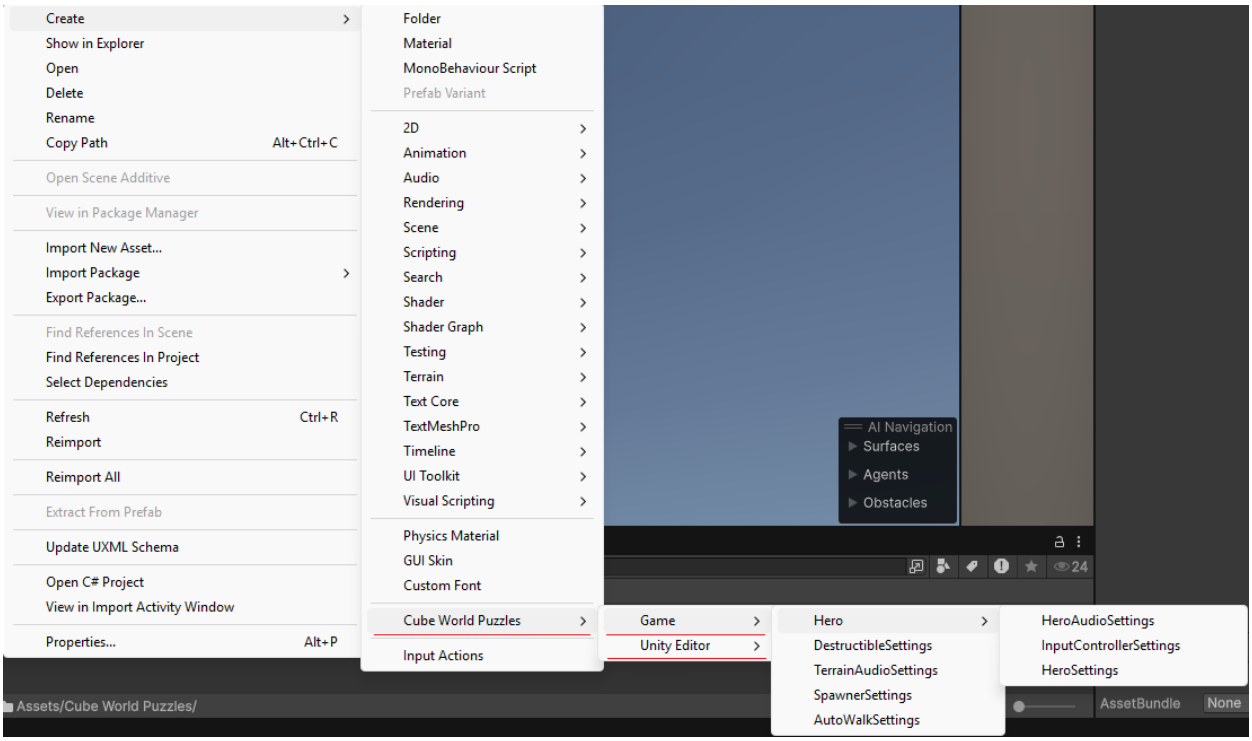

Figure 3-Create new scriptable objects

TerrainManager.cs

Terrain is the main controller in this system. All obstacles' details are storing and managing in this part. All audio clips and win event are handling from this part. The system automatically searches for this script and in cases where it fails to find it, you are offered to create it completely automatically with just one click or add it through the menu. After the user requests, everything proceeds automatically. All events are designed on left and right click in unity editor window. Select each item or use Ctrl + left click to select multi-items.

Path: Cube Worlds Puzzles\Scripts\Cube World\Environment\Terrain\TerrainManager.cs

Here we have some general options. Each section has its own variables.

Table 2-Terrain manager tab

Tab name	Description
Layout	All obstacle layout materials are store here. It helps AI find and use the right materials when building a game environment.
Local settings	This material(s) help designer to make it`s own design materials same time at AI design.
Create terrain	Environment build settings
Audio	Audio store
Information	Some more information

Layout

All obstacle layout materials are store here. It`s a scriptable object mode setting that needs to be created and defined by the user. If the user does not create this variable, it will be loaded by default from the settings created in the Resources folder.

Path: Cube Worlds Puzzles\Resources\Settings\TerrainLayout.asset

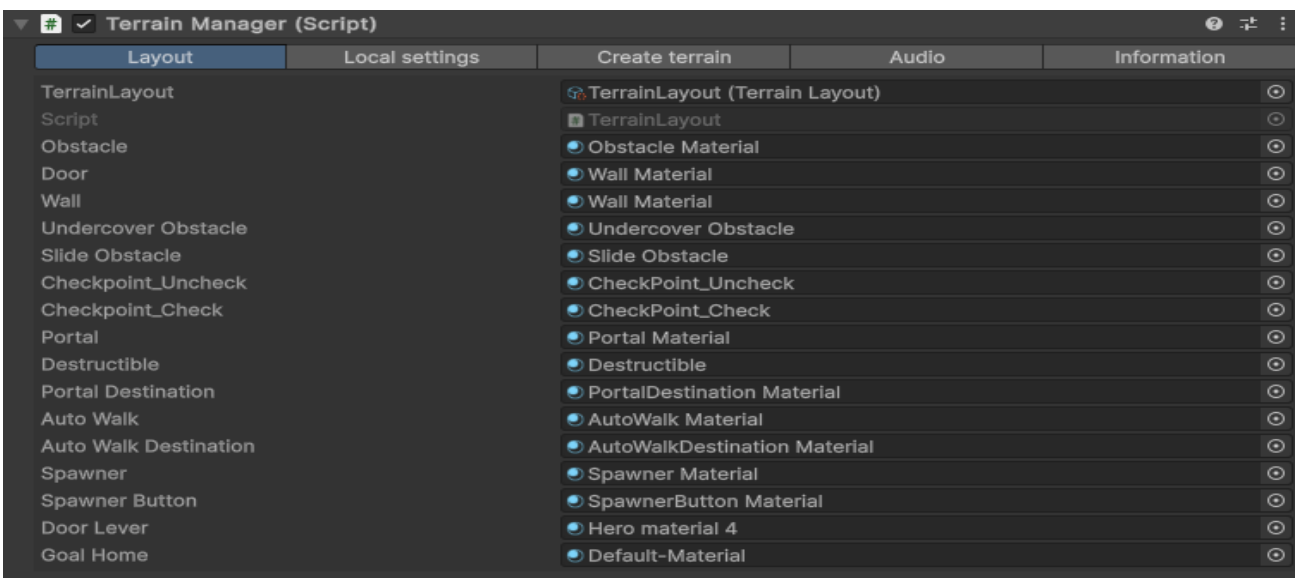

Figure 4-Terrain Manager Layout tab

Local settings

In this section, the user can manually introduce their desired material to the system for it`s own design.

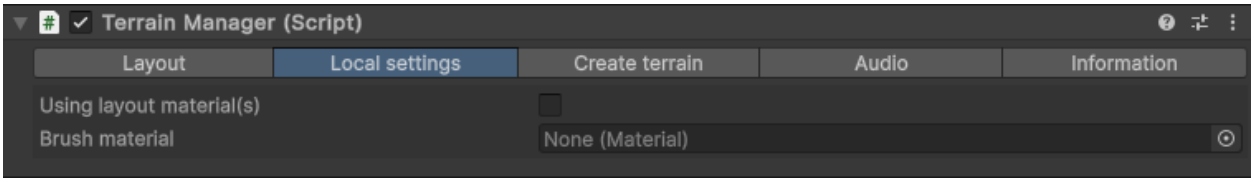

Figure 5-Terrain manager Local settings tab

Create terrain

This section provides the user with options for automatically designing the terrain. Further, by right-clicking on any of the designed elements in the scene, the user is offered even more options.

The “add first obstacle” make the first obstacle in the scene. The terrain contains the number of obstacles that each obstacle has it`s own behavior and events.

The “optimize cube” make all obstacle optimization in rendering. It removes and merges redundant vectors, which helps reduce memory space and speed up rendering.

The “Quadrilateral” creates a random playground, which will be explained in more detail in the section on playground construction.

The “Draw land” create a very simple terrain, which will be explained in more detail in the section on playground construction.

The “Cleaning” used to free up buffer space and optimize storage space.

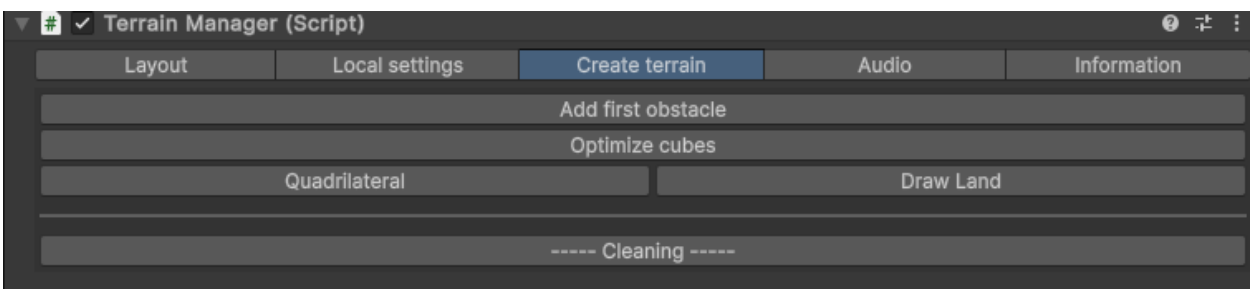

Figure 6- Terrain Manager Create terrain tab

In the figure below you can see a terrain created with the "Draw Land" feature. This option quickly creates a rectangular terrain with a desired area and with the creation of a desired hole. The yellow color means creating a hole in the terrain.

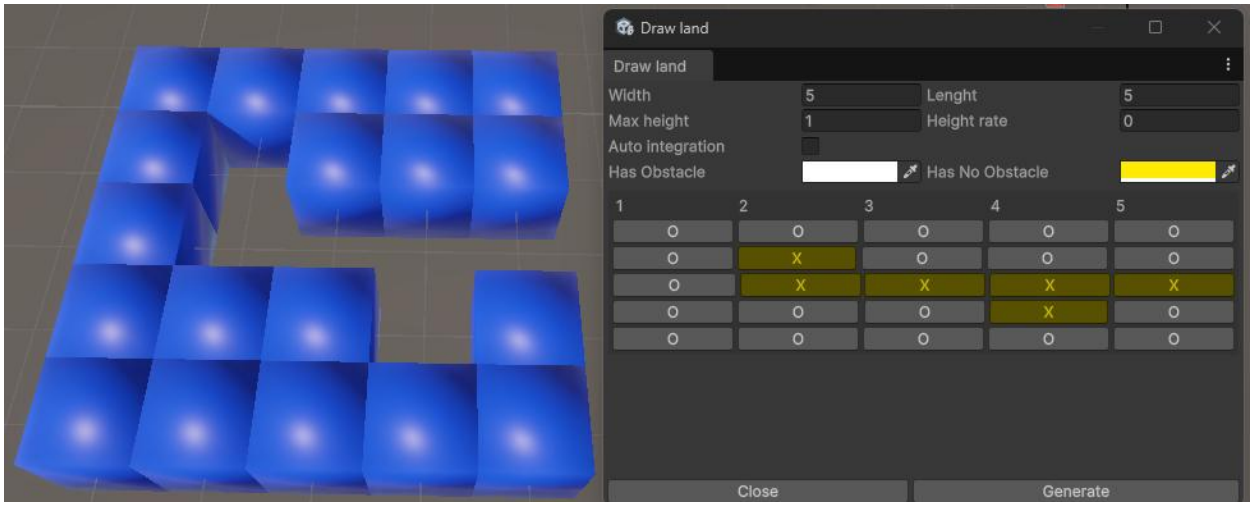

Figure 7-An example of building a land with the "Drwa land" option" option

Audio

All sounds related to terrain, including slipping, detecting traps, opening and closing doors, etc., are defined in this section. It`s a scriptable object mode setting that needs to be created and defined by the user. If the user does not create this variable, it will be loaded by default from the settings created in the Resources folder.

Path: Cube Worlds Puzzles\Resources\Settings\TerrainAudioSettings.asset

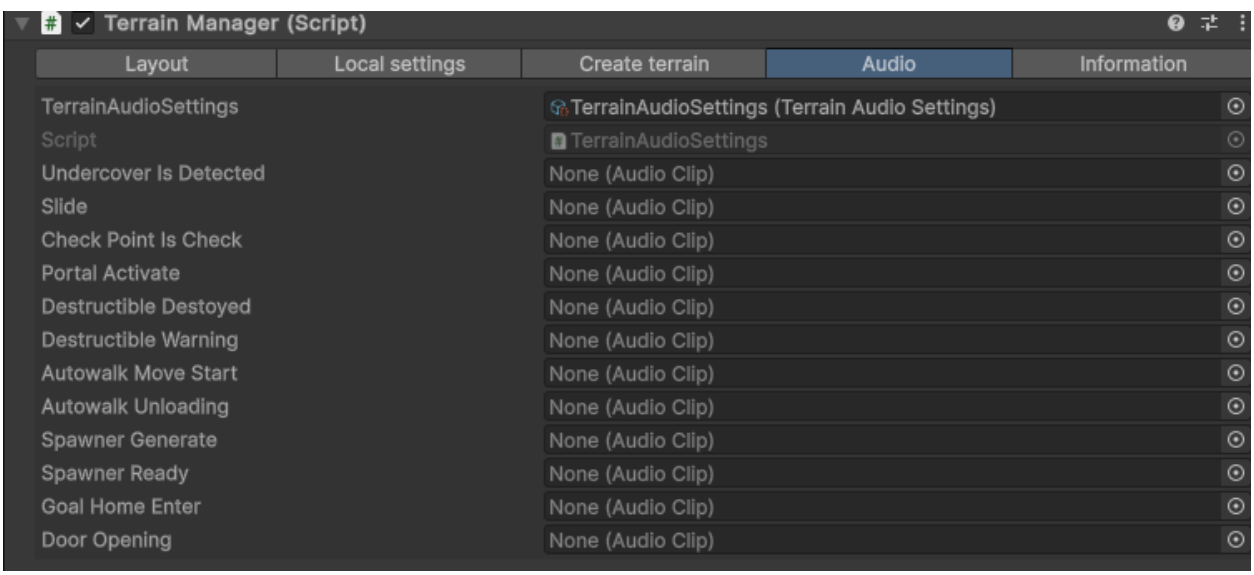

Figure 8-Terrain Manager Audio tab

HeroController.cs

This part is known as the brain controlling the protagonist of the game. An unlimited number of heroes can be added to the game and control one or more of them at the same time, which helps create more diverse stages. Below is a picture of one of our heroes, along with her environmental sensors

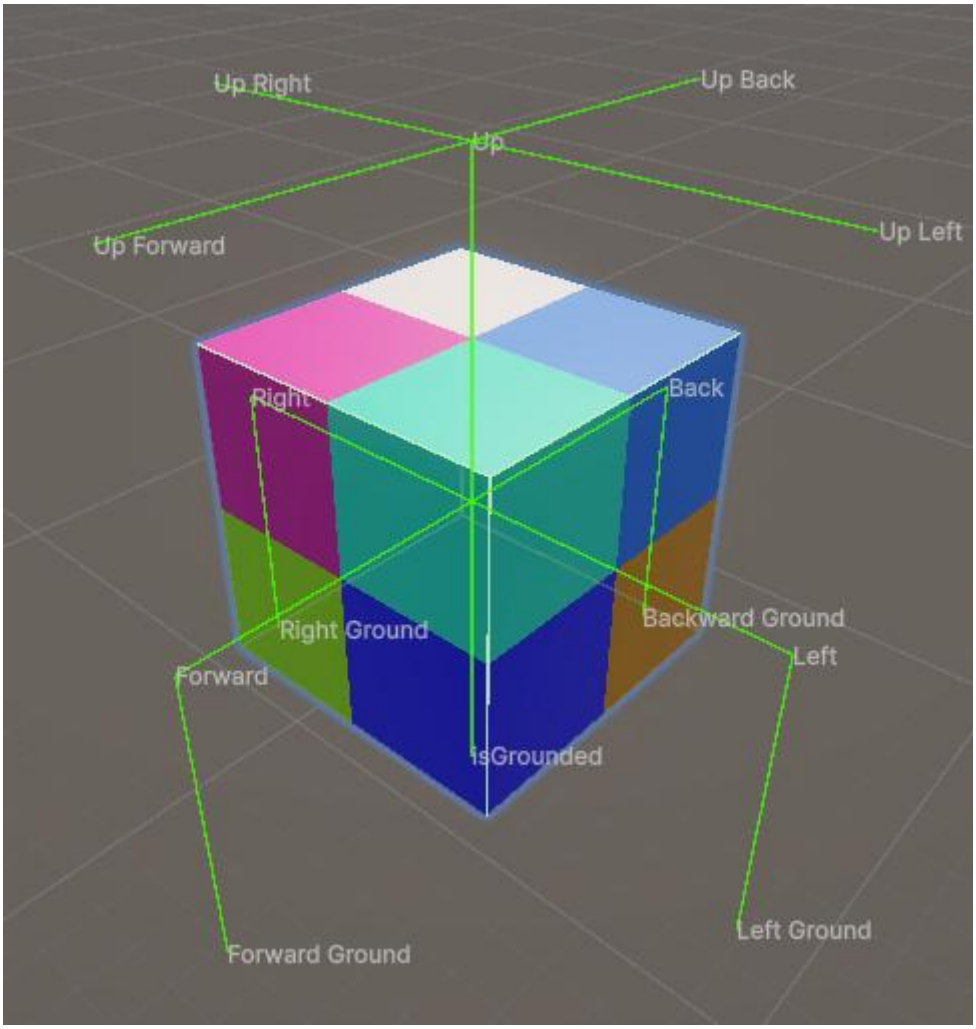

Figure 9-A picture of one of the heroes of our story

Local settings

The “Respawn able” if enabled, after the hero dies, the hero will return to the last checkpoint in the game or to the location where the game started, and the event related to the hero's death will be completed by the user in the Events tab.

The “Hero wheel” and “Hero head” are game objects that animated on hero movements.

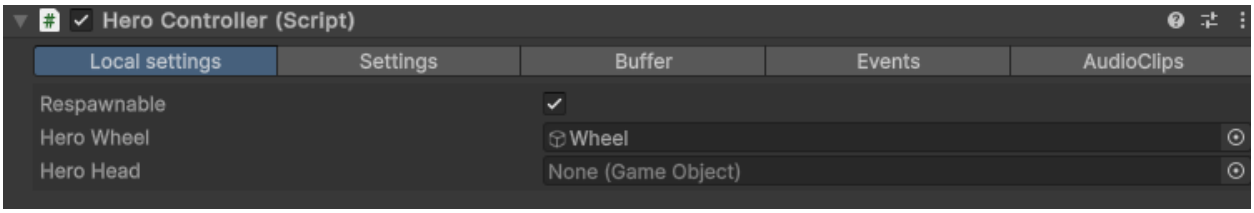

Figure 10-Hero controller Local settings tab

Settings

In this section, settings such as hero speed and environmental variables can be accessed and adjusted.

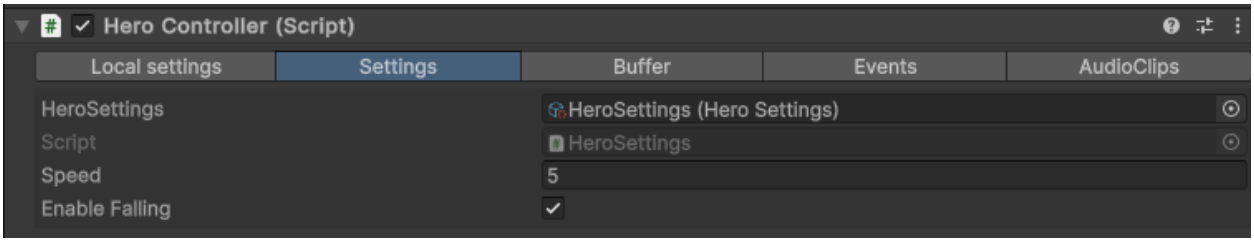

Figure 11-Hero controller Settings tab

Buffer

This section contains variables that the main algorithm shares between various sub-algorithms, including algorithms that run in specific situations. The purpose of this direct access is to allow for better understanding and to allow for testing and verification in various situations for the user.

Events

This section lists every event that may occur during the game, and the user can easily connect those moments to their functions and variables. These events are listed below:

- On hero initialize
- On hero idle
- On hero falling
- On hero movement
- On hero phase change
- On hero dead

Audio clips

All sounds related to hero, including footsteps, falling, collecting, etc., are defined in this section. It`s a scriptable object mode setting that needs to be created and defined by the user. If the user does not create this variable, it will be loaded by default from the settings created in the Resources folder.

Path: Cube Worlds Puzzles\Resources\Settings\HeroAudioSettings.asset

InputController.cs

This part is used to give commands to the hero's movement. In this version of the program, directions can be created only with the help of the keyboard and input axis. In the future, features such as joysticks and others will be added.

The “Controller type” specifies whether the hero is controlled by the user or automatically based on environmental variables.

The “Settings” is a scriptable object mode setting that needs to be created and defined by the user. If the user does not create this variable, it will be loaded by default from the settings created in the Resources folder.

Path: Cube Worlds Puzzles\Resources\Settings\InputControllerSettings.asset

The “Input type” is two deferent input type:

- Keyboard
- Input Axis

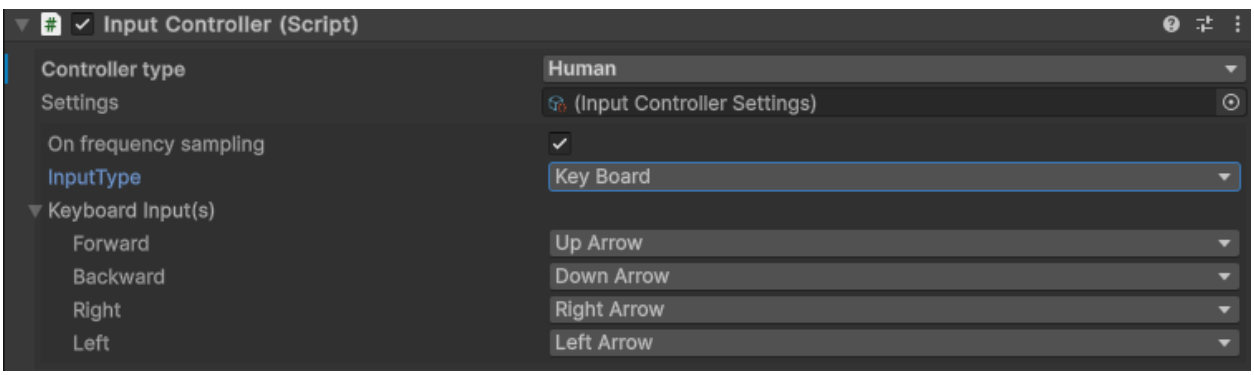

Figure 12-Input Controller

How to create and design my dreams?

First you need to create the terrain by adding from menu -read “How to use menu?” section- and then you can design your map easily. You can create first obstacle by using Create terrain tab in “TerrainManger.cs” section or right click on unity editor scene and choose “create first obstacle node”. Each terrain including from numbers of obstacles that each obstacle has their own behavior and application.

Obstacle

Obstacle is the smallest unit in environmental management. In this system, obstacle is custom-made for optimal design and better performance. You can select a single obstacle with the left mouse button, and you can select a specific number by holding down the control key. If you right-click on one or more selected obstacle(s), a menu of options will appear.

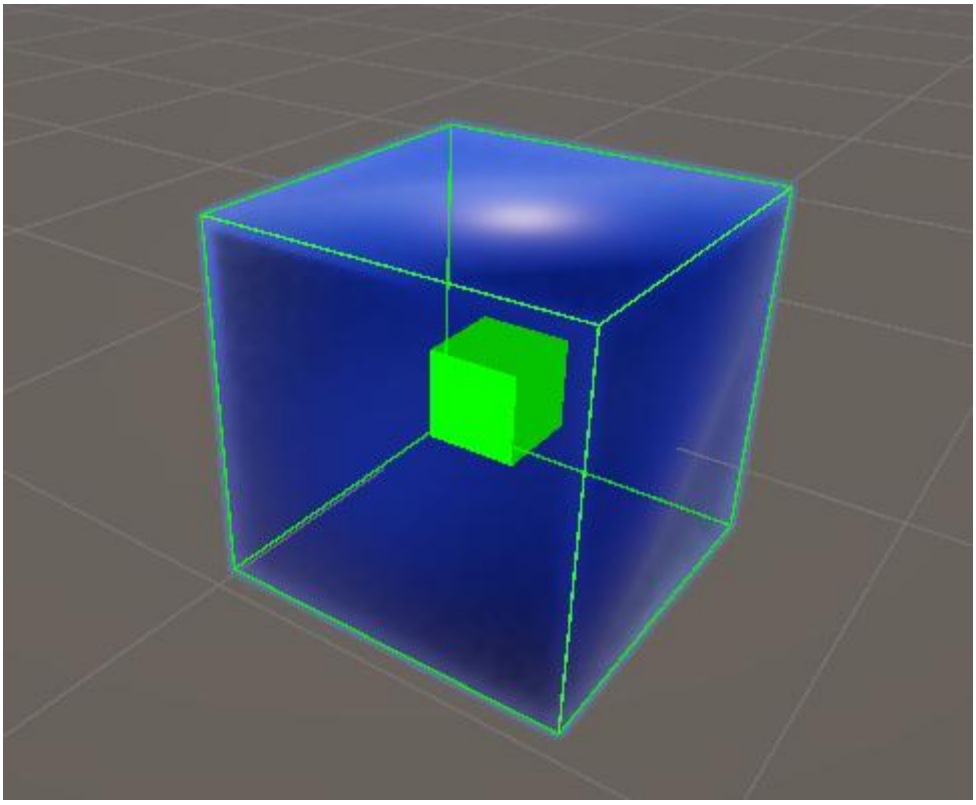

Figure 13-A picture of selected obstacle

In fact, your design starts with the creation of the first emir and you can expand it endlessly. Right-clicking on each obstacle will display a menu similar to the one below, explaining each of its features.

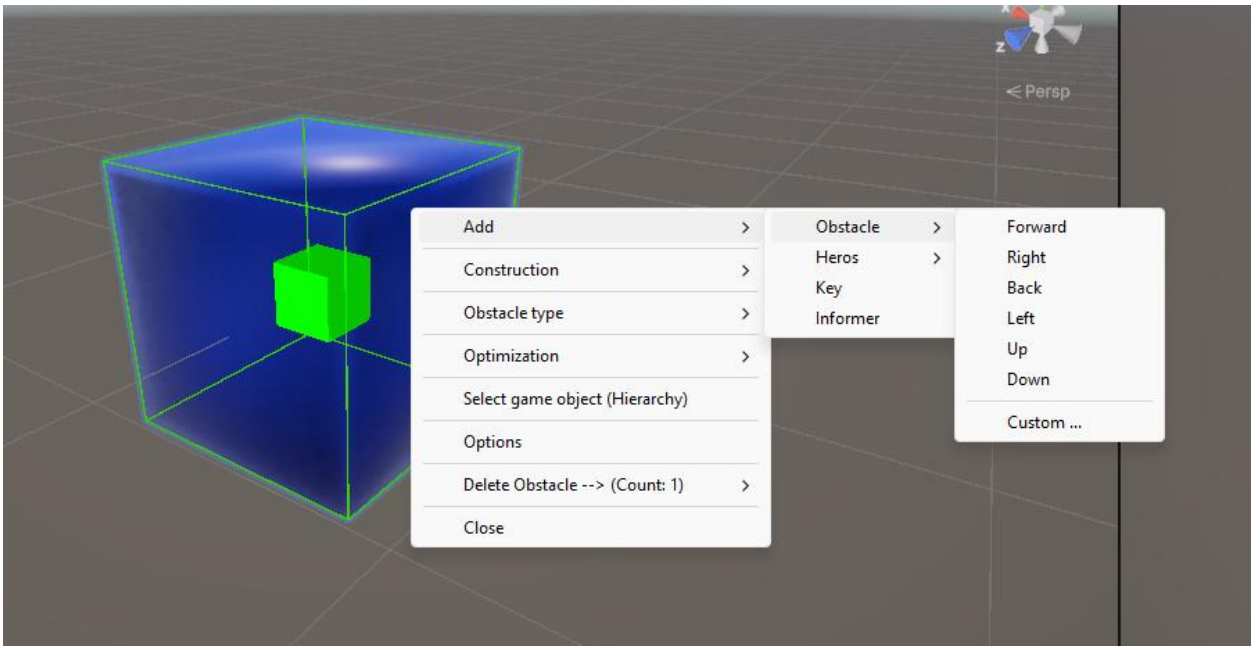

Figure 14-Obstacle menu - Add

The Add option is used to add anything to the scene. You can add obstacles to the scene in different directions as desired, or you can create an endless expanse by customizing. Don't worry if you don't get a good connection with this method, because by left-clicking on any obstacle you can expand it as far as the mouse moves.

You can load the desired number of heroes on the page by selecting the hero option. "Keys"are used to open doors, which will be discussed in detail later. “Informer” is also used to display a text to the user when the hero is hovered over its icon.

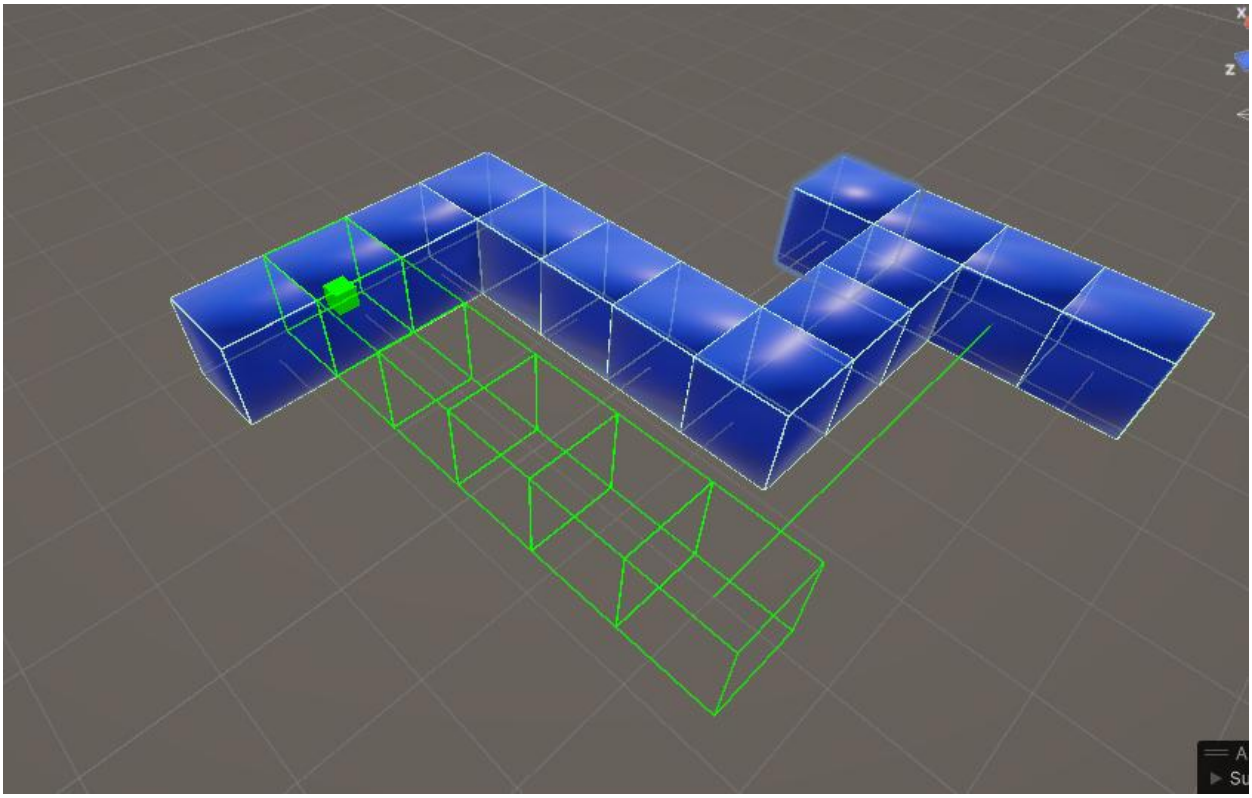

Figure 15-Obstacle free create mode

As you can see, the auxiliary lines also help the user to draw better. This procedure is very helpful in designing large maps with large length and width.

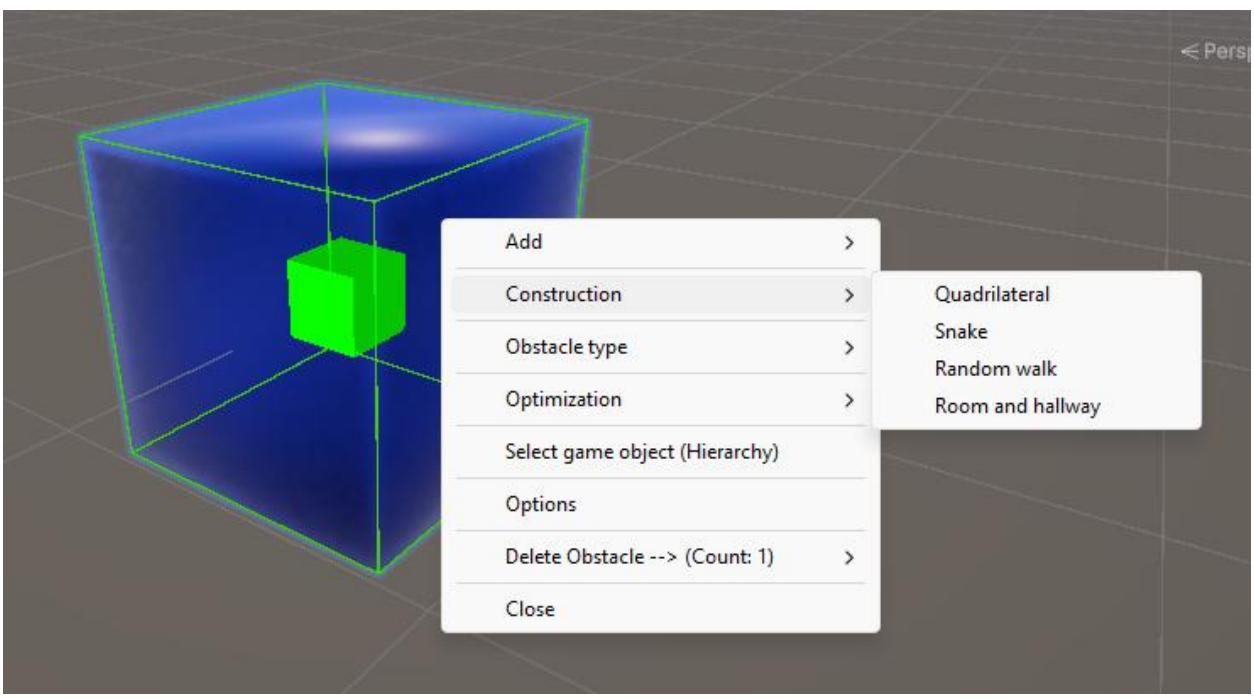

Figure 16-Construction over obstacle

Quadrilateral

This option will display the settings window where you can get a plot of land with a desired area and random heights or slopes with the help of its own algorithm. I am sure that if you try this option and combine it with other options, you can enjoy extraordinary results.

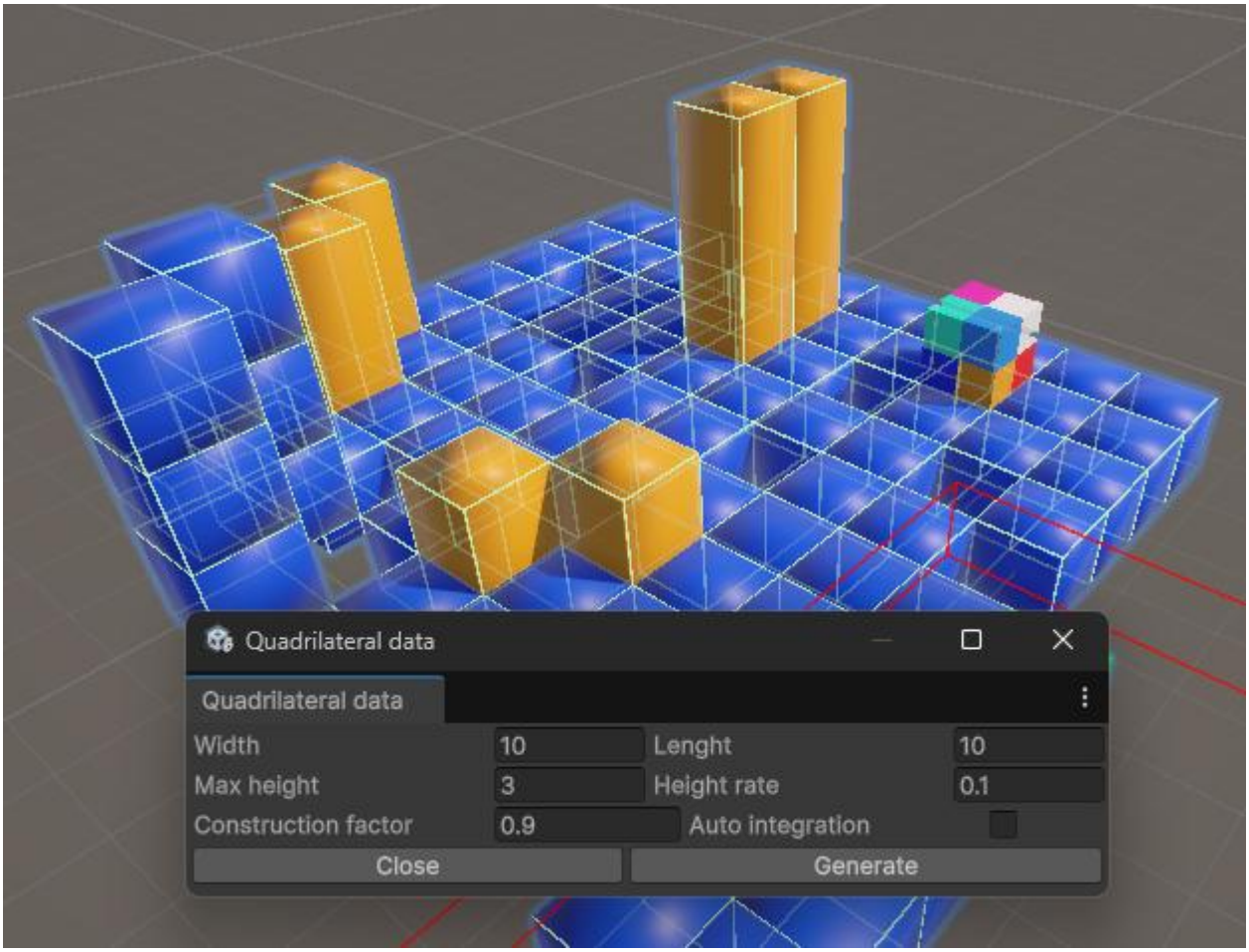

Figure 17-Quadrilateral random terrain generation

The “width” and “Length” is the length and width of a square plot form a square, where “Max height” represents the maximum height of the plot, "Height rate" represents the probability of its occurrence, and "Construction factor" is the probability of a post on the plot. The "Auto integration" option also optimizes tall columns by default.

Snake

The "Snake line" option leads to continuous development of the land in random directions. In this case, if you set the "Height rate" coefficient to zero, a flat load-bearing land is obtained.

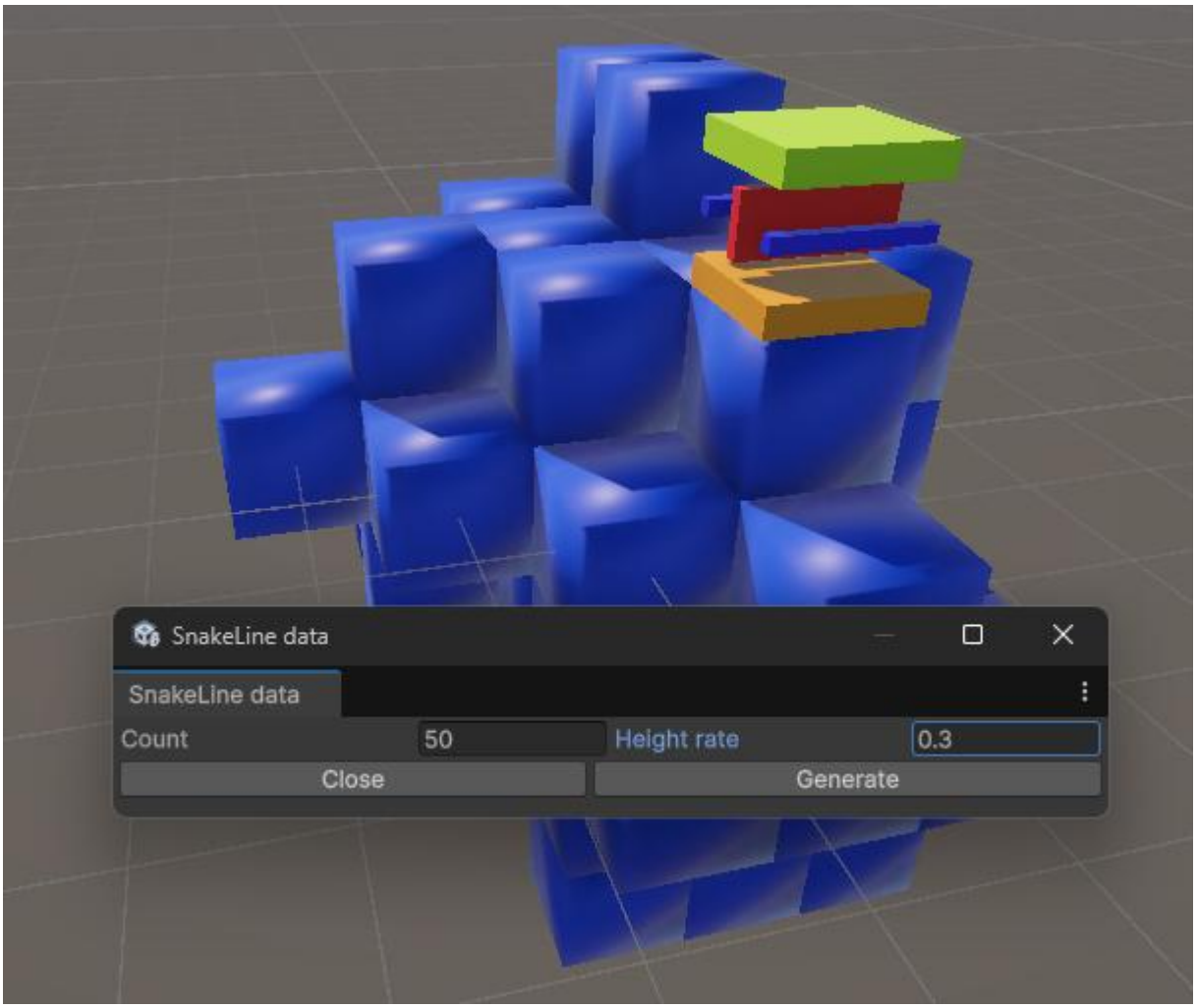

Figure 18-Snake line sample

Random Walk

The "Random Walk" option leads to the development of the terrain with the specified conditions in such a way that after determining the area of the terrain, the number of steps the algorithm takes in random directions on this terrain ends. This algorithm is very effective for creating maps similar to simple random game rooms. The image below shows an example of randomly generated terrain using this algorithm. Each time you use this feature, you can achieve a variety of structures.

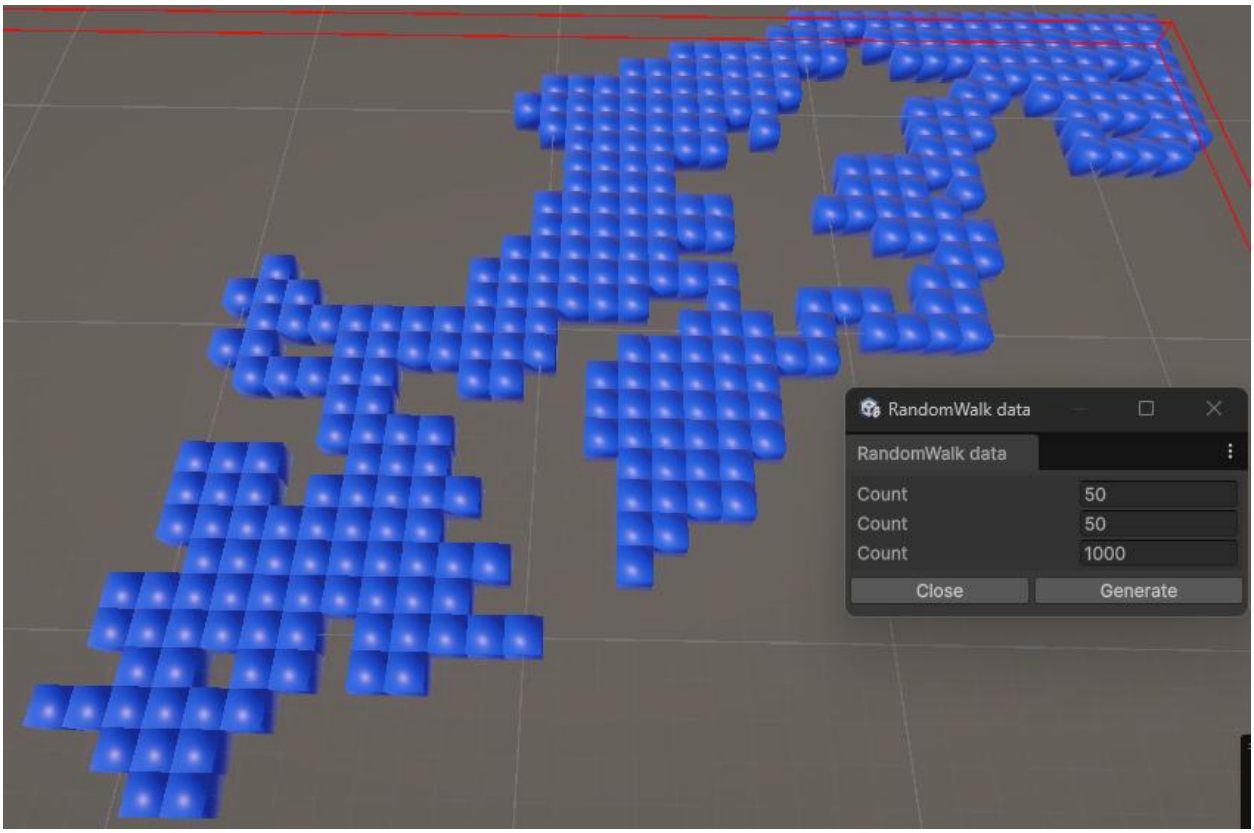

Figure 19-Random Walk sample generation

Room and hallway

This dungeon algorithm is specifically designed for building game rooms such as castles with rooms connected by corridors. One of the unique features of this algorithm is its orderliness in design while being random.

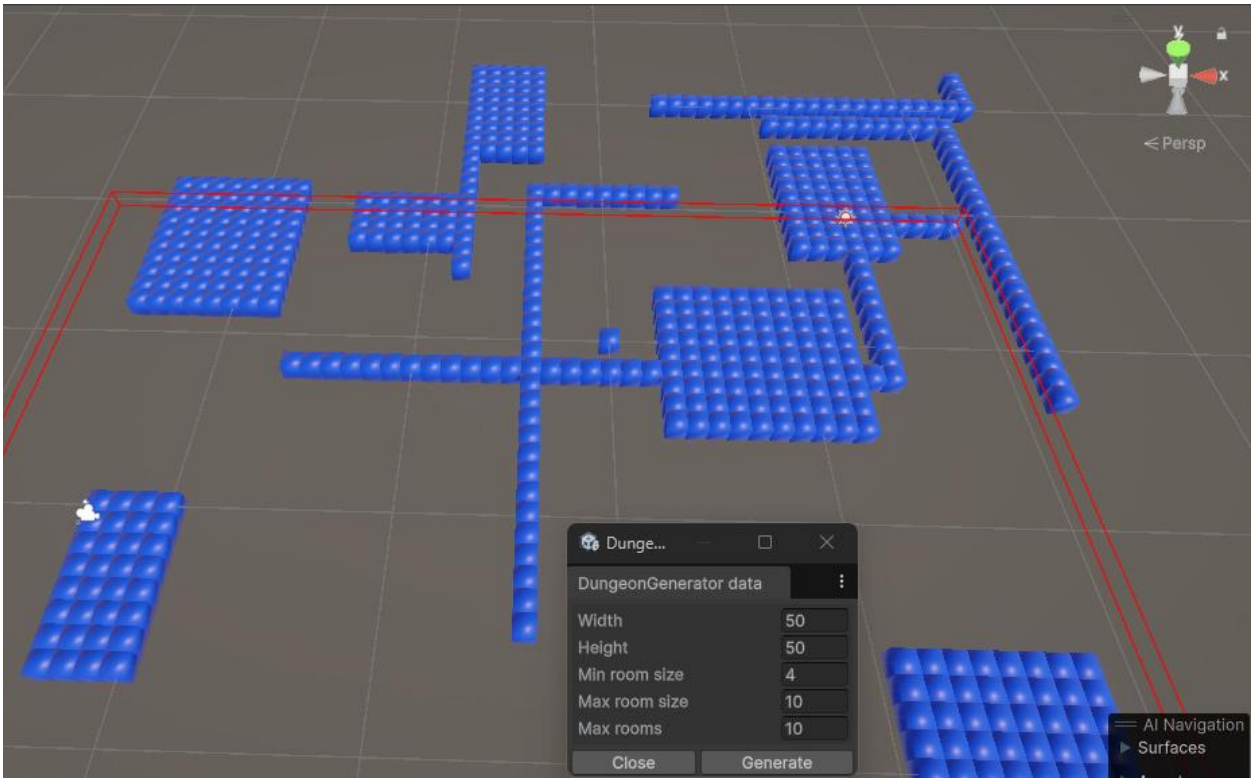

Figure 20-Room and hallway sample generation

Optimization

Finally, after fully designing your stages, you can use this option to provide various optimizations for your game scene.

The “Face rendering” is used so that you can avoid rendering one or more sides of obstacle's face.

The “Integration” is so that you can integrate multiple cubes that are connected to each other.

The “optimize all cubes” automatically calculates the various aspects of obstacles that do not appear in the camera and disables automatic rendering of that part.

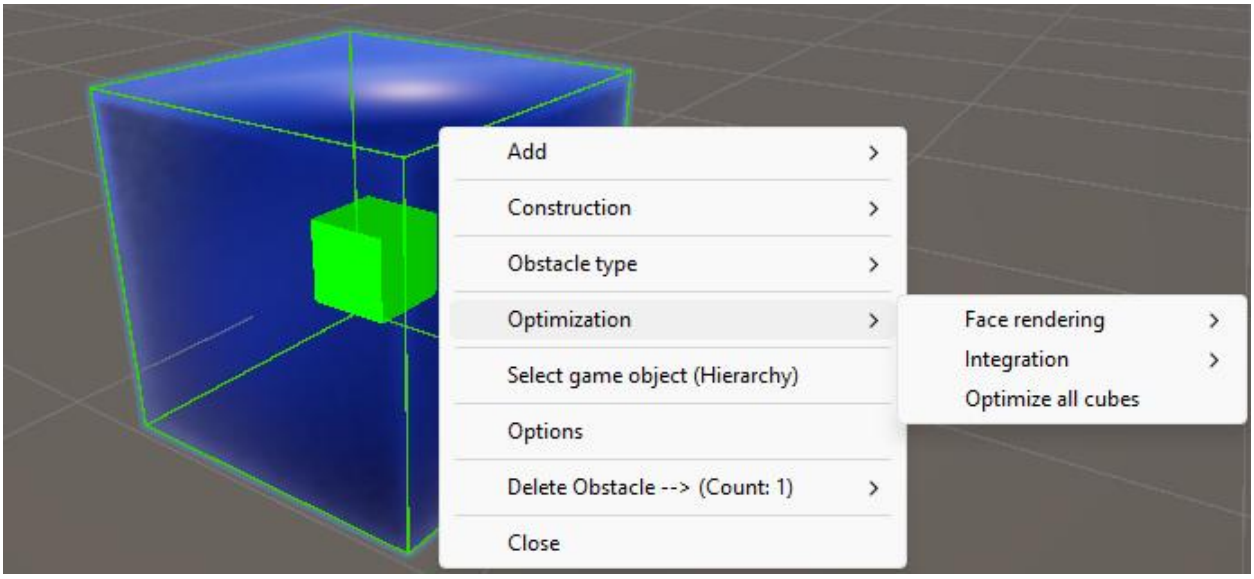

Figure 21-Obstacle’s optimization

Door

When multiple obstacles are selected, the "Convert to Door" option is enabled, allowing you to convert them into doors. In this system, doors can only be opened with a key or a lever, which can be adjusted by the game designer and play an important role in solving the puzzle. The hero cannot pass through a door that has not been opened. The system automatically checks and announces the possibility of not building a door. Doors must be square cubes or rectangular cubes.

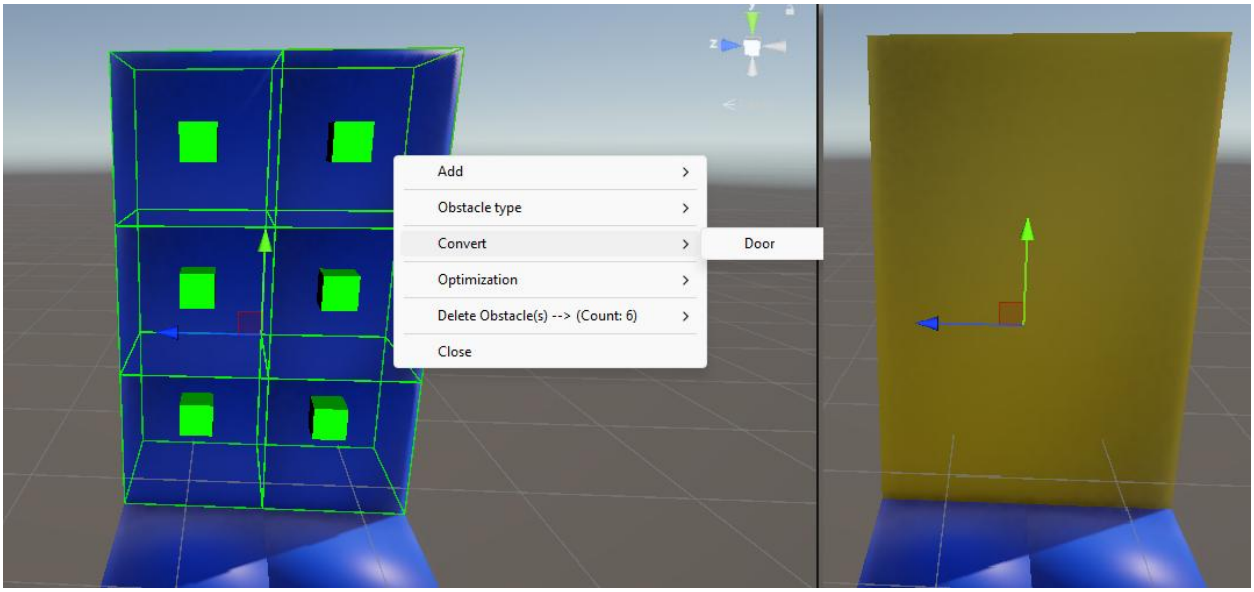

Figure 22-Converting obstacles to Door

There are three types of doors:

- None: like a wall, the hero can't get past it in any way.
- Lever open: Only when the hero stands on the "obstacle type: Door lever" will the selected door or doors begin to open. All settings related to this section can be accessed through the " Door lever " settings.
- Key open: It can only be opened with a certain number of keys.

The “Settings” include door opening and closing speed.

The “Events” include door events.

The “open key count” is count of kay that required to open the door.

The “current key count” is that count of keys that door is used.

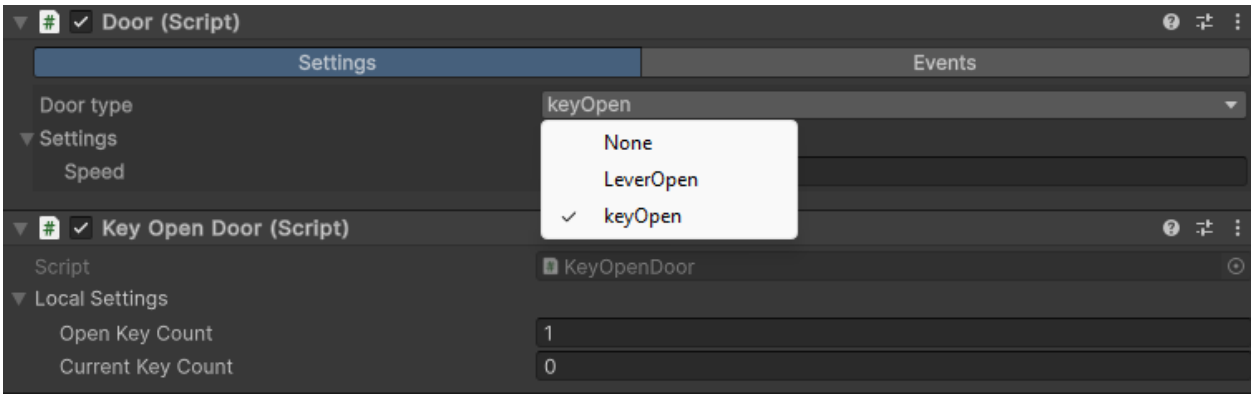

Figure 23-Door settings

Obstacle Type

Depending on the type of obstacle defined, it can have its own characteristics and behaviors, which is very important in the design of puzzles and the gameplay. In all cases, the user can add their desired event from the events window. This system has considerable dynamics.

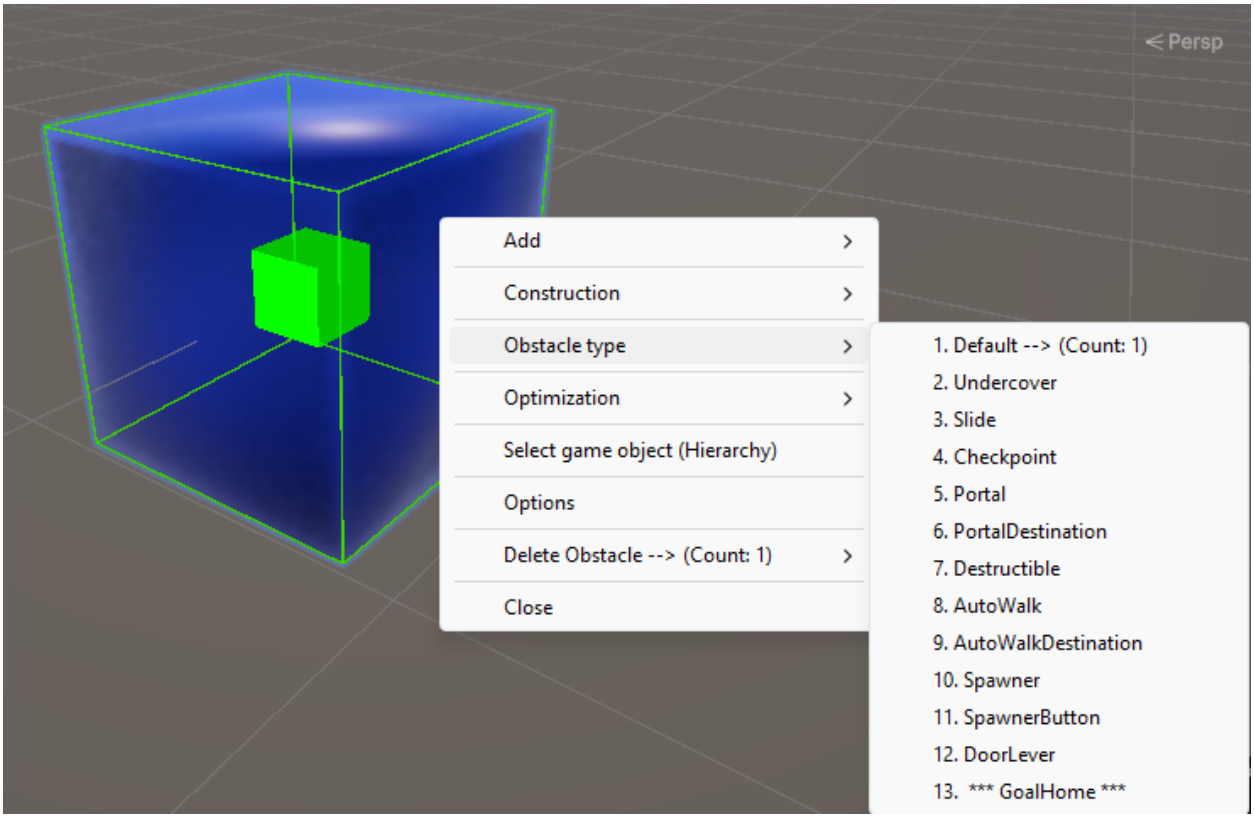

Figure 24-Obstacle types

Default

This is the default type that does not show any behavior unless the user adds a function to the event header.

Undercover

This type is just like the "Emir" mode, except that it is initially hidden and only appears when the user steps on it.

Slide

In this type, when the hero steps on it, it starts to slip.

Checkpoint

This type is self-explanatory. It is for a point to save the game. I personally use this point as part of the main element of solving complex puzzles. Therefore, the user can develop an exciting game by defining the events that occur in this type.

Portal

Portals are used to transport the hero to a desired location. I mostly use them to enter closed rooms, high up in the mountains, and remote areas. The portal can be given an "Portal destination" as a destination or a transfer.

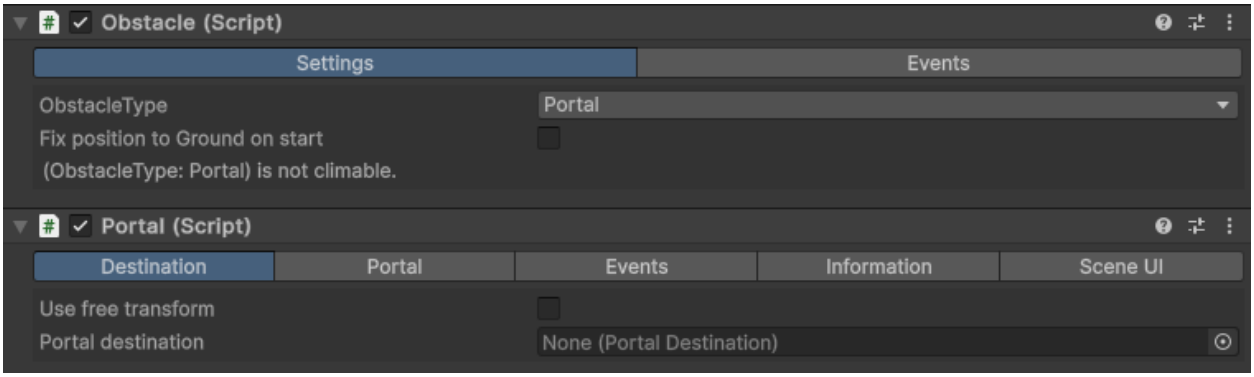

Figure 25-Portal settings

Degradable is designed to be destroyed after a limited number of passes over it.

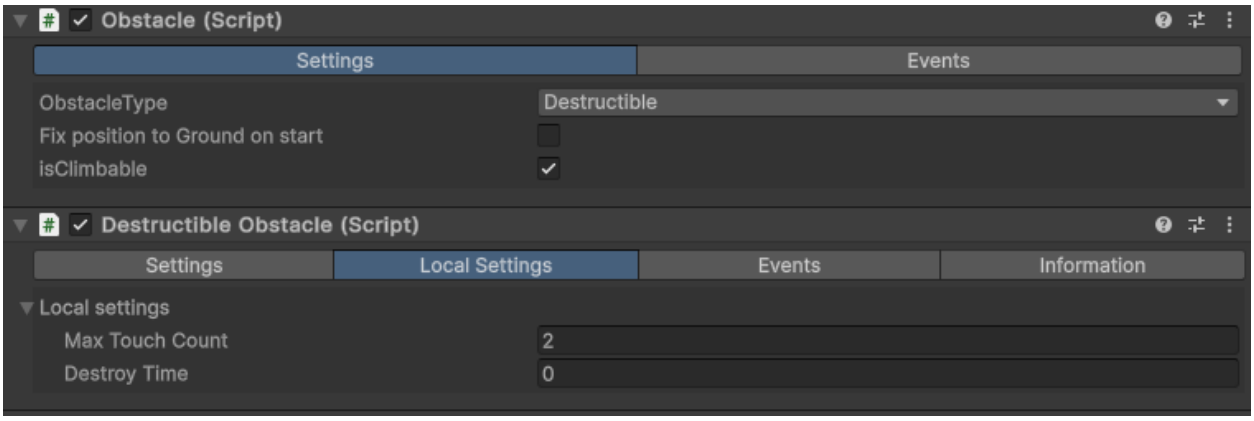

Figure 26-Destructible settings

Auto walk

Auto walk can be considered a portal for hero movement, but this is in the form of a lift and elevator with more animation.

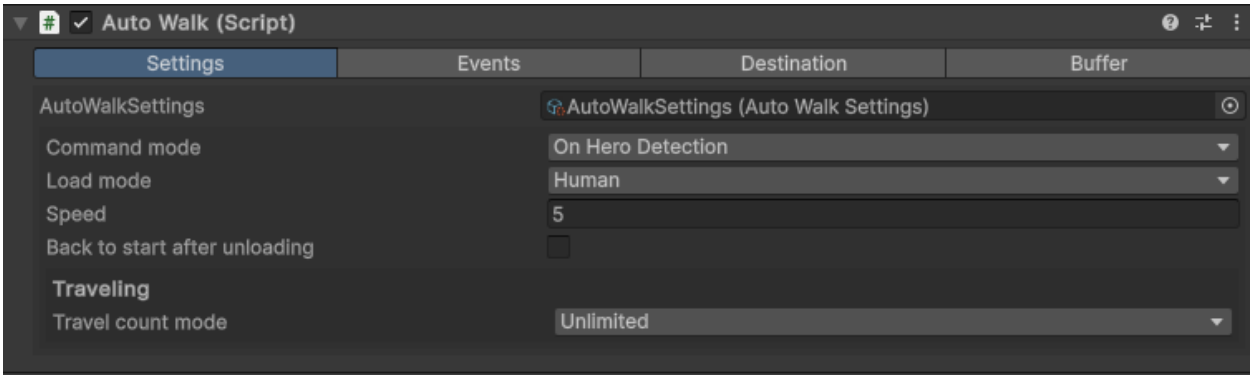

Figure 27-Auto Walk settings

Spawner

The spawner is used to produce a new hero, either under user control or as a helper in the game.

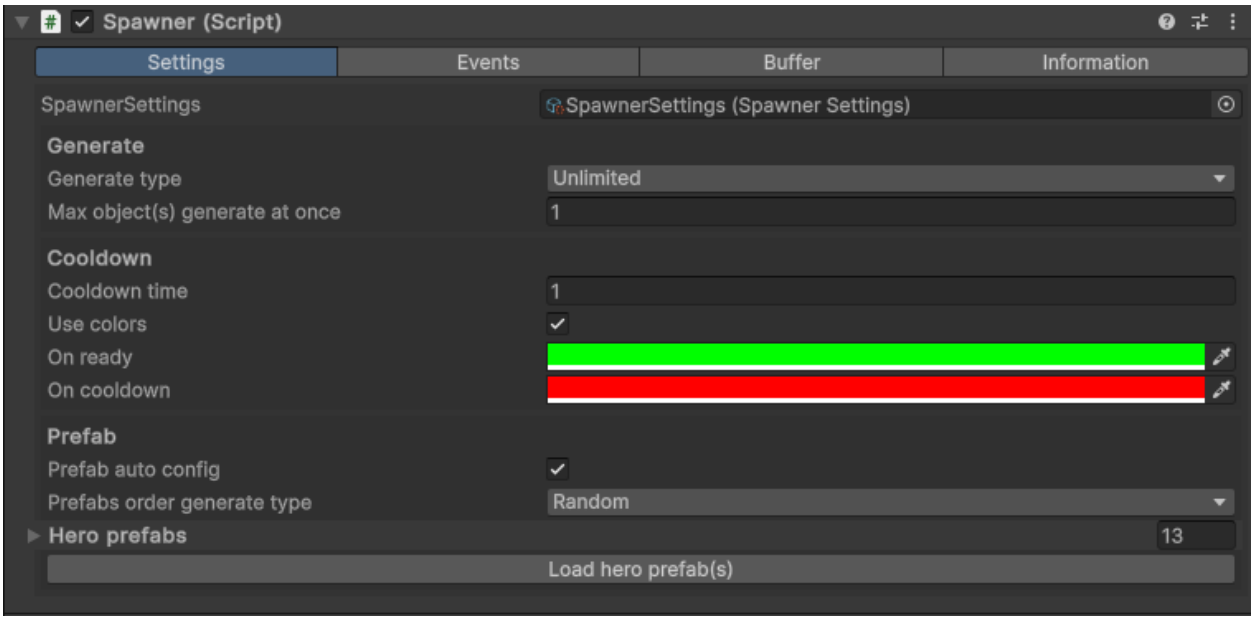

Figure 28-Spawner settings

Spawner button

The spawner button is responsible for transmitting the production order to the spawner. When the hero is placed on this button, the production order is given to the spawner.

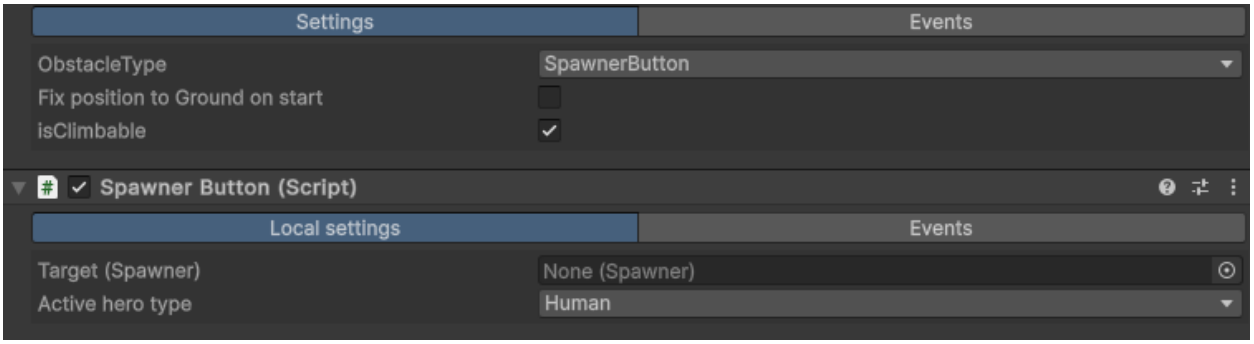

Figure 29-Spawner button

Door lever

The door lever is used to transmit the command to open or close the door. Of course, the door must also be of the lever type.

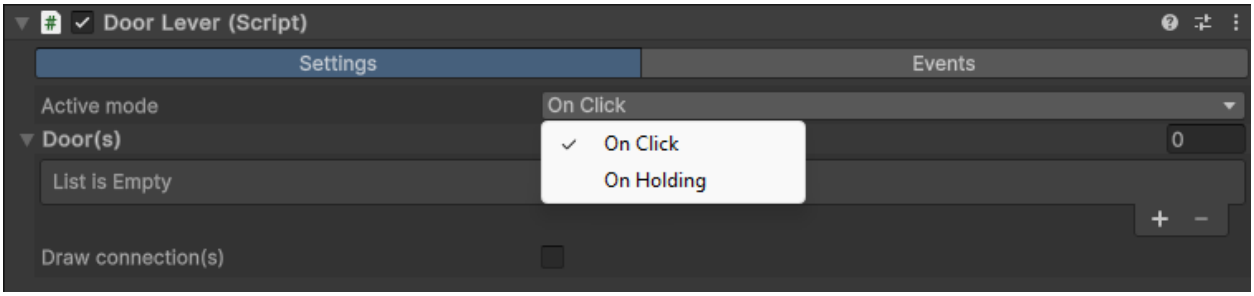

Figure 30-Door lever

Goal home

The target house is considered the end of the game. When the number of target houses you defined is reached, a hero is placed on each of them.

Contact us

You need help? Or want to improve this code?

If you need any further assistance, please contact us.

Email: hashgame.cs@gmail.com
YouTube: Hash Game Studio - YouTube